CHAPTER 1

THE PROBLEM AND REVIEW OF LITERATURE

Background of the Study

Every day, many students, workers, and citizens engage in traveling long distances from Cavite to Metro Manila. For many people, this act goes beyond just being part of their routine and becomes a stressful experience. There are many reasons why people encounter issues during their travel, such as crowded buses, long wait times, floods, dangerous terminals, high prices, and unpredictable traffic conditions.

This problem is not limited to merely posing an inconvenience. The transportation system in Metro Manila has had its fair share of inequality problems due to poor infrastructure despite the growing number of passengers. This becomes more pressing for those commuting from Cavite since the area experiences significant urbanization and population increases. What might seem like a safe road to traverse in the morning can transform into a flooding disaster by the evening, depending on whether it is raining. In terms of safety, what were meant to be safe spaces can become sources of anxiety. Additionally, any changes made to pricing can affect students and underprivileged workers financially.

Present navigation tools such as Google Maps and Waze can be useful indeed, but they were built for use by people who use automobiles. While they can determine the fastest

route that someone can take using a car, they fail to address several concerns about the trip for public transportation passengers, such as the actual cost of the trip; road flood; safety at certain terminals; and possible effects if some roads are closed and the need to explore other options. These are some factors that make commuters depend on what they already know.

As technology continues to advance, there is a huge chance to fill these gaps in the present systems. An intelligent commuting tool, which will combine the traffic data, weather alerts, flood alerts, safety information, and cost information, will greatly influence how Caviteños and Manilenos travel on a daily basis.

LakbayRoute, which stands for “route optimization application that takes into consideration various factors,” became an apt solution to the problem faced by commuters in Cavite and Metro Manila. The application is referred to as a Smart Commute Recommendation and Risk Analysis System with Traffic, Weather, Flood, Fare, and Safety-aware Route Optimization for Commuters in Cavite and Metro Manila. It is a cross-platform mobile and web application designed to give commuters the right information at the right time via one interface only. The use of travel preferences, Fastest, Cheapest, or Safest, combined with real-time data provided by reliable third-party APIs, helps make each journey easier and safer.

Review of Related Literature and Studies

The development of the LakbayRoute is based on a wide research base which sort of includes smart mobility, intelligent transport systems that use artificial intelligence

technology, GIS, flood detection, fare optimization, and also software quality assessment. Some of the literature on this research topic from 2020 to 2026 will be described below. But, really the focus of the discussion is more thematic than chronological, so it won't follow a strict timeline.

Smart Mobility and Urban Transportation Challenges in the Philippines

Commuting in the cities of the Philippines, especially Metro Manila, is already recognized not as just an annoyance, but as a solid problem in a way that sticks. In the words of the Center for Popular Democracy Group (CPDG, 2025) , the daily commute in Metro Manila is an effort defined by inequality , with infrastructure being repeatedly proven insufficient to handle the demand of millions of people who rely on public transportation services every day. These conclusions match the findings from research done by Ado, Kitano, and Morimoto (2024), where they looked at smart mobility indicators used in Metro Manila and noticed that global standards do not really mirror local realities of the transportation infrastructure in the cities of the Philippines. It became clear in their study that approaches rooted in local evidence are not optional anymore for the commuting population.

The approach taken by Domingo et al. (2021) leans on citizens' experiences, showing that smart city solutions work better when the focus is placed on commuter experiences. The building of a citizen-centered framework inside a mobile application, intended to promote inclusion among commuters, also highlighted how digital technologies can help connect people and make daily life better. And this is sort of the core idea on which the design concept for the LakbayRoute mobile application is based, you know, directly.

AI and Intelligent Transportation Systems

Over the past decades, artificial intelligence has shown its immense potential in transport planning and management. For instance, Dikshit et al. (2023) analyzed the application of intelligent routing based on AI and showed how such a system could optimize vehicle movement, thereby helping reduce traffic congestion. Furthermore, Mancino (2022) explored the role of autonomous cars and AI for creating effective smart transport systems that improve traffic management and routing capabilities beyond conventional systems.

Vaidya, Kulkarni, and Didore (2021) offered an overview of intelligent transport systems that are created based on IoT technologies. As they note, the introduction of sensors and other data-generating devices allows for developing highly adaptive transportation platforms, thus making them efficient and flexible. The use of AI for managing transportation systems was also discussed by Shatnawi, Al-Omari, and Al-Qudah (2020). Specifically, they showed how geographic information systems could be used together with AI to optimize bus stops' location in such networks. Among the works presented in this section, one stands out because it deals directly with the problem faced by the proposed system. Alizadeh et al. (2021) conducted a feasibility study of how the flood mapping of cities based on traffic signs could help optimize routes. Thus, this study shows the technical feasibility of integrating flood risks into routing algorithms and, consequently, is directly relevant to the development of the flood risk assessment module of LakbayRoute.

Weather-Aware and Climate-Resilient Transportation Planning

The weather, especially when it's rainy, plus the whole flood situation, really ends up being one of the most important factors but somehow it still gets poorly mixed into a lot of modern navigation apps. In his study about AI use for urban transport systems design, Liu (2025) points out that transportation networks should be able to react to weather shifts in a kind of live way. This then surfaces one of the biggest gaps you can see in many navigation applications, which is just… treating weather as if it does not matter when suggesting routes.

Kikuchi (2025) continues that idea by doing a comparison evaluation between weather aware AI algorithms and the more traditional route optimization AI algorithms. The results are pretty straightforward, they show that including weather does give a noticeable jump in productivity while people are traveling. This is not just theoretical either, it becomes crucial in places like Cavite and Metro Manila because disruptions show up due to typhoons almost every year. So yeah, the value of weaving weather conditions into navigation systems cannot really be overstated. Because of that, LakbayRoute uses an integrated weather system that draws from PAGASA and OpenWeather APIs.

Flood Monitoring and Detection Systems

Flood is one of the most disturbing and threatening problems for commuters living in the low-lying parts of Cavite and Metro Manila. According to AWARE Monitoring Systems (2025), every city requires sophisticated flood monitoring infrastructure, which could help not only city administrators but common citizens too by giving them advance notice about upcoming flooding. The authors proved the feasibility of incorporating such technologies into transportation planning, thus showing how to use such information in reality.

An example of a very important local study would be the IMPROVE Floodeye system, proposed by Dita (2025). This system includes a mobile application with predictive routing based on ensemble learning algorithms and is designed to predict which routes could be flooded and warn drivers about the danger. Thus, Floodeye might be considered among the closest analogues to the future LakbayRoute. Nonetheless, it should be pointed out that, similarly to other flood-monitoring solutions, this system operates at the stage of detecting flooding without any consideration of fares and passenger safety.

Fare Optimization and Public Transportation Cost Management

For a lot of travelers, like students and daily wage earners, the transport fare issue cannot be treated as secondary. Transport fare actually sits in their daily expense, and, in a way it becomes the core, not some side detail. The problem about price-optimal routing in public transport systems was tackled by Euler and Borndörfer (2022). They built a mathematical model aimed at finding the most economical routes inside multi-modal transit networks. Their work suggests that fare data should not just be presented as information, but it should be really used as an input when choosing routes. LakbayRoute carries this principle by using a fare estimation tool, it relies on the latest approved fare tables for jeepneys, buses, and UV Express.

Software Quality Evaluation Using ISO/IEC 25010

However, the evaluation of software that will be used for public purposes needs to follow a certain structure and framework in order to provide reliable results. The ISO/IEC 25010 software quality model is considered to be one of the most effective models of

software evaluation. The use of ISO/IEC 25010 has been discussed in Codacy (2025), in which the usefulness of using this software quality model in evaluating Functional Suitability, Performance Efficiency, Usability, and Reliability was explained. Moreover, Monterail (2026) described how the implementation of ISO 25010 models in software development processes could contribute to decreasing technical debts.

Synthesis

From all the available literature, there seems to be a glaring void in the realm of transport technology options available for Filipino commuters. On one hand, it has been observed that intelligent route planning systems employing AI algorithms show great promise when it comes to traffic management purposes; on the other, weather/flood monitoring tools appear capable of detecting and tracking various environmental hazards. Yet, despite these promising results, it seems as if no existing system especially in the case of commuting between Cavite and Metro Manila has managed to bring all of these elements into a cohesive whole where traffic knowledge, weather and flood detection, individual commuter-oriented fares and safety considerations are all taken into account in one recommendation tool. This is exactly the kind of hole that LakbayRoute is meant to fill..

Statement of the Problem

Commuters moving along the Cavite-Metro Manila route experience a wide range of problems during their trips such as traffic jams, floods, incidents on roads, risks of travel within transit terminals, and high prices of transport service. At the same time, modern GPS systems provide route recommendations but they are tailored to the needs of drivers using

personal vehicles while they do not take into account issues important to public transport passengers fares, floodable areas, terminal security, and risks of travel. This problem calls for designing a system that will be focused on commuters' requirements.

The study aims at developing the application named "Lakbay Route" (Smart Commute Recommendation and Risk Analysis).

This study seeks to explore the following research problems:

- In what way can an AI-based system make use of real-time information regarding traffic conditions, weather, flooding, fare rates, and safety considerations in order to generate optimal routing advice for travelers moving in the Cavite-Metro Manila axis?
- What mechanisms can the system use to compute and show the estimated fares of jeepneys, buses, and other PUVs based on the revised fare matrix of the LTFRB
- How does the system take note of flood-prone areas and dangerous terminals among other road safety threats in making routing decisions?
- How can the system provide commuters with real-time alerts about traffic incidents, road closures, flooding, accidents, and other factors that can affect their trip?
- How acceptable is the system in terms of Functional Suitability, Performance Efficiency, Usability, and Reliability in line with the ISO/IEC 25010 software quality model?

Objectives of the Study

General Objective

The major objective of the research will be the development of the LakbayRoute system, which uses artificial intelligence to provide Smart Commute Recommendation and Risk Analysis, thus delivering more efficient and affordable routes based on up-to-date data such as traffic, weather, flood hazards, fare, and safety.

Specific Objectives

The objectives of this research include:

- The design and implementation of an AI-based route suggestion system that examines the traffic situation, costs of travel, risk of floods, and safety issues and recommends routes based on the user's chosen travel preference: Fastest, Cheapest, or Safest.
- The development of a fare estimator tool that calculates the cost of taking public transport via jeepneys, buses, and public utility vehicles based on up-to-date LTFRB approved fare tables.
- The inclusion of weather and flood detection features using third-party API services (e.g., PAGASA and OpenWeather) to determine flood-prone places and issue travel warnings.
- The implementation of a hazard assessment module that detects dangerous stations, areas with high incidents of accidents, and sections with flooded roads that could affect commuters in their travels.

- The design and implementation of an alert system for traffic jam, road closures, accidents, floods, and other disruptions in transport.
- The assessment of the quality and acceptability of the implemented system through pilot-testing using the ISO/IEC 25010 software quality model, specifically the Functional Suitability, Performance Efficiency, Usability, and Reliability aspects.

Conceptual Framework

The Input Process Output (IPO) model forms the basis for understanding the process through which data flows in order to give appropriate commuter directions. It defines the system process into three consecutive steps:

Input: Input data for the system include various types of data, namely user preferences such as Fastest, Cheapest, or Safest mode, real-time traffic data acquired from the Google Maps API, weather and rain data collected from PAGASA and OpenWeather API, fare rate matrices approved by the LTFRB for jeepney, bus, and UV Express fares, flood data (both real-time and historical data), and data regarding safety such as incident reporting and identification of risky terminals.

Process: The process involves the integration of inputs through an AI-powered route analysis module that evaluates the potential routes on the basis of a number of factors such as risks, delays, and efficiency considerations. This process includes computing the cost of transportation based on the LTFRB approved tariff, evaluating the route by means of

parameters such as traffic delays, flooding, and safety risks, and providing alerts concerning traffic incidents, flooding, accidents, and road closures.

Outputs: The outputs include proposed routes based on user preferences, expected time and cost of travel, notifications related to disruptions, warnings about safety threats at dangerous terminals and road sections, and a user-friendly interface for mobile and web applications.

IPO Model LakbayRoute AI is an intelligent commuting advisor that transforms raw transportation information into useful and practical recommendations.

Significance of the Study

The design of LakbayRoute is based on the needs of the commuter population. The following are some of the potential beneficiaries of LakbayRoute in this study:

Commuters

Among the millions who commute daily from Cavite to Metro Manila, the development of LakbayRoute provides an important tool for making decisions on their journey based on facts rather than experience or guesswork. This makes it easier to make use of the available routes while keeping in mind factors like the safety of the route, the estimated time of travel, and cost of the trip; all these help minimize their stress.

Students/Office Workers

Commuting students and office workers to educational and work institutions will greatly benefit from this application during adverse travel conditions like heavy traffic,

sudden rains, or flooded streets. This helps them to make better choices and plan their journey according to their strict schedule.

Local Government Units (LGUs)

Information provided through LakbayRoute, including the frequency of travel conditions that include flooding in some roads, dangerous locations where most accidents happen, and unsafe terminals, is useful to LGUs regarding transport-related issues.

Transportation Industry

In contributing to the overall goal of improving public transport, the study helps in creating awareness of how the use of data and technology can be done responsibly. In allowing passengers to make more informed decisions, the usage of transportation systems is maximized because of LakbayRoute.

To Future Researchers and Developers

In the field of AI-enabled transportation systems, smart mobility, and route optimization, the study serves not only as a reference but also as a framework for future studies that will be conducted in the Philippines.

To The Researchers

As researchers and students, this research gives us a chance to use our knowledge and skills in artificial intelligence, programming, API implementation, database manipulation, and system design into practice.

Scope and Delimitation

Scope of the Study

- The present research revolves around the development of the cross-platform mobile and web-based LakbayRoute that is meant to give recommendations and analyses of risks related to commutes from Cavite to Metro Manila. The study covers the following points:
- Creation of artificial intelligence-based route recommendations for users of public transportations such as jeepneys, buses, and UV Express with a focus on selected routes in both Cavite and Metro Manila.
- Provision of the estimation of fares as well as prediction of travel time based on the latest approved fare matrices provided by the LTFRB and existing routes.
- Implementation of the traffic observation and analysis with Google Maps Platform.
- Implementation of the weather and flood observation with data provided by PAGASA and OpenWeather APIs.
- Analysis of travel risk due to identification of dangerous terminals, hazardous locations, and flood-risky areas.
- Real-time notifications for traffic congestion, road blocks, floods, accidents, and other issues.
- Evaluation of the system in terms of Functional Suitability, Performance Efficiency, Usability, and Reliability through a questionnaire designed in compliance with the requirements of ISO/IEC 25010.

- Collection of evaluation data from at least thirty (30) respondents including regular commuters, students, and employees who will be found using the purposive sampling technique with PITX as the main venue.

Delimitations

LakbayRoute is designed to give practical assistance to travelers, subject to the following limitations:

- Data from Google Maps, PAGASA, and OpenWeather are crucial to the system's efficiency. Errors or outages from any third-party provider may hamper the performance of the app.
- Internet connection is required, and users with slow connections on their end may experience problems.
- Fast-changing developments, such as sudden road blockage, flood, and traffic accidents, are not always instantaneously incorporated into the system due to the reporting process.
- Information about traffic and the number of people at terminals cannot be fully realized without available transportation data from relevant offices.
- The scope of application is limited to the Cavite-Metro Manila area and beyond; it will not cover other places or help drivers navigate using personal cars.

Definition of Terms

The following is how these terms are used in this study:

Artificial Intelligence (AI)

The term AI refers to the methods and algorithms involved in enabling the program to process multivariate transportation information, recognize patterns, and recommend appropriate routes based on user preferences and current situation.

Fare Matrix

An approved table of fares from the Land Transportation Franchising and Regulatory Board (LTFRB). It will be used in this study to calculate accurate estimates of costs in using jeepney, bus, and UV Express.

Flood Risk

Probability of a certain road or area becoming inaccessible or hazardous because of flooding, which will be calculated in this study based on the use of current weather data and past instances of flooding.

ISO/IEC 25010

International standard in software product quality which gives eight quality characteristics upon which software products can be evaluated. The evaluation of software quality will focus on four of the eight in this study: functional suitability, performance efficiency, usability, and reliability.

LTFRB (Land Transportation Franchising and Regulatory Board)

Philippine government agency responsible for the regulation of the public transport industry and the setting of fares. Approved LTFRB fare matrices will serve as the basis for the fare calculator function of LakbayRoute.

Multi-Criteria Decision Making (MCDM)

A decision-support tool for making judgments concerning alternative options from several criteria. MCDM will be used in the research to assess and prioritize the commutation routes based on speed, cost, safety, traffic, and weather conditions.

PAGASA (Philippine Atmospheric, Geophysical and Astronomical Services Administration)

The national meteorological organization of the Philippines. PAGASA provides the main source of weather and flood alerts used in this study.

Public Utility Vehicle (PUV)

A passenger vehicle franchise issued by the government to provide shared transport using modes of transport such as jeepney, bus, and UV Express vans. PUVs form some of the transportation modes targeted by the LakbayRoute application.

Real Time Data

Data that has been collected, processed, and transmitted with minimal delay. Real-time data used in this study include traffic information, weather reports, and alerts on floods and road closures.

Route Optimization

This entails finding the most optimal route between two points based on specific criteria. Route optimization is carried out on LakbayRoute application based on the three user preferences, namely, Fastest, Cheapest, and Safest routes.

Risk Assessment

The procedure of assessing potential dangers that may pose a threat on the commuting route. Flood-prone roads, accident-prone sites, and dangerous terminal areas form part of the risk analysis conducted in this study to generate warnings.

UV Express

This refers to point-to-point public utility vehicles governed by the LTFRB which utilizes shuttles and vans as the main means of transport. UV express forms some of the primary transportation modes considered by the LakbayRoute application.

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